

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 13: Motion and Time

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	13 — Motion and Time
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors weaponise the plain-mean average-speed error over multi-leg journeys, the inverted km/h $\leftrightarrow$ m/s factor, slope-vs-y-value graph confusion, rest/idle time dropped from the denominator, cumulative-graph descent misread as extra distance, the T proportional to $\sqrt{L}$ pendulum scaling, and overtaking/crossing reasoning on distance-time lines. Do not write on this paper — mark answers on your answer sheet.

1. A courier rides the first 60 km of a straight route at 40 km/h, then the next 90 km at 60 km/h. The average speed for the whole 150 km run is:

- (A) 52.5 km/h (B) 48 km/h
(C) 50 km/h (D) 100 km/h

2. A cyclist covers the first one-fifth of a straight path at 6 km/h and the remaining four-fifths at 24 km/h, all five parts equal in length. The average speed for the whole path is:

- (A) 15 km/h (B) 12 km/h
(C) 18 km/h (D) 20 km/h

3. A falcon stoops at 270 km/h. Expressed in metres per second this is:

- (A) 972 m/s (B) 7.5 m/s
(C) 75 m/s (D) 97.2 m/s

4. A high-speed shuttle moves at 35 m/s. In kilometres per hour this is:

- (A) 9.72 km/h (B) 126 km/h
(C) 63 km/h (D) 175 km/h

- 5.** A hiker walks for 2 hours at 5 km/h, rests for 1 hour, then walks for 1 hour at 9 km/h on a straight trail. The average speed for the whole outing, including the rest, is:
- (A) 7 km/h (B) 6.33 km/h
(C) 9.5 km/h (D) 4.75 km/h
- 6.** Two pendulums hang from one beam. Pendulum P has a 4 m string and a heavy steel bob; pendulum Q has a 1 m string and a light wooden bob released from a wider angle. The ratio of P's time period to Q's time period is:
- (A) 2 : 1 (B) 4 : 1
(C) 1 : 2 (D) 1 : 4
- 7.** A dashboard has a needle that climbs and falls with the accelerator, steady at the '54' mark, and a window of digits that had advanced by 162 since the trip began. If the trip's average matched that 54 km/h reading, the time it has lasted so far is:
- (A) 3 hours (B) 108 hours
(C) 1 hour (D) 0.33 hours
- 8.** A swift flies at 25 m/s and a train runs at 81 km/h. After converting to a common unit, the faster one and its speed in km/h is:
- (A) the train, at 81 km/h, since 25 m/s is only 6.9 km/h
(B) they tie at 90 km/h (C) the swift, at 108 km/h
(D) the swift, at 90 km/h
- 9.** A beetle crawls 30 cm in 5 s, pauses 6 s, then crawls a further 45 cm in 10 s along a straight line. Its average speed over the whole observation, in m/s, is:
- (A) 0.0357 m/s (B) 0.05 m/s
(C) 0.075 m/s (D) 0.0625 m/s
- 10.** On a distance-time graph an object's line slopes from (0 s, 0 m) to (3 s, 36 m), runs flat from (3 s, 36 m) to (9 s, 36 m), then slopes from (9 s, 36 m) to (12 s, 90 m). Its speed during the steepest segment is:
- (A) 12 m/s, during the first segment (B) 36 m/s, read off the flat middle
(C) 18 m/s, during the last segment (D) 7.5 m/s, averaged over the whole 12 s
- 11.** A train covers 30 km in the 1st half-hour, 45 km in the next half-hour, and 60 km in the third half-hour along a straight track. The most precise true statement is:
- (A) the motion is uniform, because the half-hour intervals are equal
(B) the motion is non-uniform and slowing down
(C) the motion is circular, because trains follow curved rails
(D) the motion is non-uniform and speeding up, since unequal distances are covered in equal times
- 12.** A sewing-machine needle plunges straight down and back up through the cloth 240 times a minute at a steady rate. The best description, and the time for one stroke, are:

- (A) oscillatory and periodic, with one stroke taking 4 s
- (B) oscillatory and periodic, with one stroke taking 0.25 s
- (C) circular, with one stroke taking 0.25 s
- (D) rectilinear and non-periodic, so no fixed stroke time

13. On a distance-time graph, line P runs from (0,0) to (6 s, 84 m) and line Q runs from (0,0) to (6 s, 48 m), both straight. P's speed exceeds Q's by:

- (A) 36 m/s
- (B) 2 m/s
- (C) 14 m/s
- (D) 6 m/s

14. A motorist wants a 360 km trip to average 90 km/h. The first 270 km were driven at 108 km/h. The constant speed needed over the remaining 90 km to just hit the planned average is:

- (A) 72 km/h
- (B) 90 km/h
- (C) 60 km/h
- (D) 120 km/h

15. A pendulum completes 30 oscillations in 75 s. Its string is shortened until its time period becomes exactly two-fifths of the original. How many oscillations does it now complete in 75 s?

- (A) 75
- (B) 30
- (C) 12
- (D) 60

16. Among these four speeds, the single greatest is:

- (A) 100 km/h
- (B) 1620 m/min
- (C) 30 m/s
- (D) 0.4 km/min

17. A pendulum's length is increased to nine times its original value. Its time period becomes:

- (A) nine times as long
- (B) one-third as long
- (C) three times as long
- (D) the same, since length does not affect the period

18. On a distance-time graph the line is straight from (0 s, 0 m) to (5 s, 50 m), horizontal from (5 s, 50 m) to (9 s, 50 m), then straight back down from (9 s, 50 m) to (15 s, 0 m). The total path length travelled over the 15 s is:

- (A) 100 m
- (B) 50 m
- (C) 0 m
- (D) 150 m

19. An object's distance-time graph is a straight line through (4 s, 14 m) and (10 s, 41 m). Its speed is:

- (A) 3.5 m/s
- (B) 4.5 m/s
- (C) 5.5 m/s
- (D) 4.1 m/s

20. A bus runs 48 km in the 1st hour, 60 km in the 2nd hour and 72 km in the 3rd hour on a straight road. Its average speed over the three hours, and the nature of its motion, are:

- (A) 60 km/h, and the motion is uniform (B) 180 km/h, and the motion is non-uniform
(C) 55 km/h, and the motion is non-uniform
(D) 60 km/h, and the motion is non-uniform

21. A boat goes 48 km downstream in 2 h and the same 48 km upstream in 4 h. Treating the whole there-and-back run, its average speed is:

- (A) 18 km/h (B) 12 km/h
(C) 16 km/h (D) 24 km/h

22. A wall clock's minute hand sweeps once round the dial every hour, returning to the 12 each time. The two descriptors that fit the tip of the minute hand are:

- (A) oscillatory and periodic (B) circular and periodic
(C) rectilinear and periodic (D) random and circular

23. A pendulum on Earth has a time period of exactly 6 s. To shorten its period to 2 s, its length should be changed to:

- (A) one-third of its original length (B) one-ninth of its original length
(C) one-sixth of its original length (D) three times its original length

24. Spot the INCORRECT speed conversion among these four:

- (A) 75 km/h = 25 m/s (B) 54 km/h = 15 m/s
(C) 90 km/h = 25 m/s (D) 126 km/h = 35 m/s

25. On a distance-time graph two straight lines start at different heights and cross at the point (8 s, 40 m). At the instant of crossing the two objects:

- (A) are moving at the same speed
(B) are at the same distance from the start at that moment
(C) both momentarily stop (D) swap their directions of travel

26. A metronome arm ticks 90 times in 60 s at a steady beat. The time between consecutive ticks is:

- (A) 1.5 s (B) 0.67 s
(C) 90 s (D) 0.015 s

27. A car covers 120 km, then is parked for 30 minutes, then covers a further 80 km. The first 120 km took 2 h and the second 80 km took 1 h. The average speed for the entire outing, including the parked time, is:

- (A) 66.7 km/h (B) 60 km/h
(C) 57.1 km/h (D) 40 km/h

28. A sand clock (hourglass) is trusted to measure a fixed interval because:

- (A) the sand grains push each other in regular pulses
(B) the two glass bulbs swing to-and-fro like a pendulum

- (C) sunlight warms the upper bulb on a fixed schedule
- (D) the sand trickles through the narrow neck at a roughly steady rate

29. A motorbike travels at a steady 108 km/h. How many metres does it cover in 75 seconds?

- (A) 8100 m
- (B) 2250 m
- (C) 1500 m
- (D) 225 m

30. On a distance-time graph an object's line gets steadily steeper while always continuing to rise. Over this stretch the object is:

- (A) moving forward and speeding up
- (B) moving forward but slowing down
- (C) moving backward toward the start
- (D) at rest after the first bend

31. A pendulum makes 48 oscillations in 72 s. Its frequency is:

- (A) 0.67 oscillation per second
- (B) 1.5 oscillations per second
- (C) 48 oscillations per second
- (D) 72 oscillations per second

32. A car drives the first 40 km at 40 km/h, the next 40 km at 20 km/h and the last 40 km at 60 km/h. The average speed for the whole 120 km is closest to:

- (A) 40 km/h
- (B) 32.7 km/h
- (C) 36 km/h
- (D) 30 km/h

33. Light travels about 300000 km in 1 s. A radio signal to a probe takes 600 s to arrive. The probe is therefore about:

- (A) 500 km away
- (B) 180 million km away
- (C) 18 million km away
- (D) 1.8 billion km away

34. Two cars start from the same point. On a distance-time graph car A's line is a straight slant from the origin and car B's line bends upward from the origin, starting gentler than A. Which statement can be correct?

- (A) A is always ahead because a straight line always beats a curve
- (B) B is always ahead because a curve always wins
- (C) B starts slower than A but can overtake and finish ahead if its line bends steeply enough
- (D) they stay together throughout since both start at the origin

35. A runner covers 240 m in 30 s, then a further 360 m in 90 s. The average speed over the full 600 m is:

- (A) 5 m/s
- (B) 6 m/s
- (C) 4 m/s
- (D) 8 m/s

36. On a distance-time graph, line A goes from (0,0) to (10 s, 150 m) and line B goes from (0,0) to (10 s, 90 m). The ratio of A's speed to B's speed is:

- (A) 3 : 5
- (B) 150 : 90 and cannot be simplified
- (C) 5 : 3
- (D) 1 : 1 because the times match

37. On a distance-time graph the line is perfectly horizontal at a height of 60 m from $t = 5$ s to $t = 13$ s. During this stretch the object's speed is:

- (A) 60 m/s, read straight off the height
- (B) zero, because its distance does not change with time
- (C) 8 m/s, the width of the segment
- (D) very high, since a flat line means smooth fast travel

38. Two pendulums hang side by side: one is 1 m long, the other 16 m long. Over one minute, comparing the number of oscillations:

- (A) the 1 m pendulum makes four times as many, because its period is one-quarter as long
- (B) the 16 m pendulum makes more, because a longer string swings more often
- (C) they make the same number, since mass is not stated
- (D) the 16 m pendulum makes exactly 16 times as many

39. A flag flutters back and forth in gusty wind without settling into any fixed rhythm. Its motion is best described as:

- (A) oscillatory and periodic, like a clock pendulum
- (B) circular, because the flag wraps around
- (C) to-and-fro but non-periodic, since it does not repeat at fixed intervals
- (D) rectilinear, because the pole is straight

40. A car covers 150 km, idles in traffic for 30 minutes, then covers a further 90 km. The moving stretches took 1.5 h and 1 h respectively. The average speed for the whole trip, including the idle, is:

- | | |
|--------------|-------------|
| (A) 96 km/h | (B) 60 km/h |
| (C) 120 km/h | (D) 80 km/h |

41. A walker goes 4 km to a park at 8 km/h and pedals back the same 4 km at 24 km/h. The average speed for the round trip is:

- | | |
|-------------|-------------|
| (A) 16 km/h | (B) 10 km/h |
| (C) 12 km/h | (D) 14 km/h |

42. An ant covers 84 mm in 7 s. Expressed in m/s, its speed is:

- | | |
|---------------|----------------|
| (A) 0.012 m/s | (B) 12 m/s |
| (C) 0.12 m/s | (D) 0.0012 m/s |

43. A pendulum makes 40 complete oscillations in 50 s. How many complete oscillations does it make in 4 minutes at the same rate?

- | | |
|---------|---------|
| (A) 160 | (B) 200 |
| (C) 192 | (D) 240 |

44. On a distance-time graph an object's line rises from (0 s, 0 m) to (4 s, 40 m), is flat from (4 s, 40 m) to (10 s, 40 m), then rises from (10 s, 40 m) to (13 s, 100 m). The fastest

segment, and its speed, are:

- (A) the first segment, at 10 m/s
- (B) the flat middle segment, at 40 m/s
- (C) the last segment, at 20 m/s
- (D) they are all equal at 10 m/s

45. A water clock keeps time by the steady drip of water from one vessel into another. Which of these would make it UNRELIABLE compared with a sundial?

- (A) a cloudy sky overhead
- (B) nothing in this list; unlike a sundial, a water clock works at night and on cloudy days
- (C) the arrival of nighttime
- (D) the Sun being low on the horizon

46. Three messengers cover the same 18 km. P takes 45 minutes, Q takes 30 minutes and R takes 36 minutes. Ranking their speeds from fastest to slowest gives:

- (A) P, then Q, then R
- (B) Q, then R, then P
- (C) R, then Q, then P
- (D) they are all equal because the distance is the same

47. A car drives the first 60 km at 60 km/h, the next 60 km at 30 km/h and the last 60 km at 90 km/h. The average speed for the whole 180 km is closest to:

- (A) 60 km/h
- (B) 55 km/h
- (C) 45 km/h
- (D) 49 km/h

48. A high-speed train runs at 216 km/h. How far does it travel in 35 seconds?

- (A) 2100 m
- (B) 7560 m
- (C) 1050 m
- (D) 210 m

49. Which statement about distance-time graphs is FALSE?

- (A) a steeper straight line means a faster object
- (B) a horizontal line means the object is moving at a steady high speed
- (C) the slope of the line gives the speed
- (D) a straight slanted line means uniform speed

50. A swimmer covers 2.7 km in 45 minutes at a steady pace. Her average speed in m/s is:

- (A) 1 m/s
- (B) 3.6 m/s
- (C) 0.06 m/s
- (D) 60 m/s

51. A pendulum clock that keeps perfect time has its bob raised so the effective string is made shorter. From then on, compared with correct time, the clock will:

- (A) run slow, because a shorter pendulum swings less often
- (B) keep correct time, since period depends only on the bob's mass
- (C) stop, because a short pendulum cannot swing
- (D) run fast, because a shorter pendulum has a shorter period and so ticks more often

52. On a distance-time graph (distance measured from the start) an object's line climbs to a peak at (12 s, 72 m) and then falls back to the time axis at (20 s, 0 m). This means the object:

- (A) kept speeding up the entire time
- (B) moved away to 72 m and then returned to its starting point
- (C) moved in a circle and vanished
- (D) stopped permanently at the 72 m peak

53. On a distance-time graph object M's straight line passes through (2 s, 16 m) and (8 s, 52 m). Its speed is:

- (A) 8 m/s
- (B) 5 m/s
- (C) 6.5 m/s
- (D) 6 m/s

54. A walker sets out at 9:00 toward a town along a road at a steady 5 km/h. A cyclist leaves the same point along the same road at 11:00 and rides at 15 km/h. The cyclist catches up with the walker at:

- (A) 1:00, after the cyclist has ridden 30 km
- (B) 11:30, after the cyclist has ridden 7.5 km
- (C) they never meet
- (D) 12:00, after the cyclist has ridden 15 km

55. A train 120 m long passes a stationary signal post completely in 6 s. Its speed is:

- (A) 720 m/s
- (B) 5 m/s
- (C) 0.05 m/s
- (D) 20 m/s

56. Two cars start from the same town. On a distance-time graph car A's straight line from the origin is steeper than car B's straight line from the origin. After any time has elapsed:

- (A) car B is always farther because its line is gentler
- (B) they stay exactly the same distance apart at all times
- (C) car A is always farther from the town than car B
- (D) car A is behind car B until their lines cross later

57. Spot the INCORRECT speed conversion among these:

- (A) 117 km/h = 30 m/s
- (B) 36 km/h = 10 m/s
- (C) 72 km/h = 20 m/s
- (D) 144 km/h = 40 m/s

58. An odometer reads 36,200 km at the start of a drive and 36,470 km at the end; the drive lasted 4 hours including a 30 minute break. The car's average speed for the whole drive was:

- (A) 77.1 km/h
- (B) 60 km/h
- (C) 54 km/h
- (D) 67.5 km/h

59. On a distance-time graph an object's straight line passes through (3 s, 21 m) and (11 s, 77 m). Its speed is:

- (A) 7.7 m/s
- (B) 6 m/s
- (C) 8 m/s
- (D) 7 m/s

60. A table records an object's distance from start as time (s): 0, 2, 4, 6, 8 and distance (m): 0, 12, 24, 30, 30. The object's motion over the 8 s is best described as:

- (A) uniform at 6 m/s for the whole 8 s
- (B) speeding up throughout the 8 s
- (C) at rest the whole time, since it ends where a flat run begins
- (D) uniform at 6 m/s for the first 4 s, then slower, then at rest in the last 2 s

Solutions — Science (Class 7), Chapter 13:

Motion and Time — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	C	31	A	41	C	51	D
2	A	12	B	22	B	32	B	42	A	52	B
3	C	13	D	23	B	33	B	43	C	53	D
4	B	14	C	24	A	34	C	44	C	54	D
5	D	15	A	25	B	35	A	45	B	55	D
6	A	16	C	26	B	36	C	46	B	56	C
7	A	17	C	27	C	37	B	47	D	57	A
8	D	18	A	28	D	38	A	48	A	58	D
9	A	19	B	29	B	39	C	49	B	59	D
10	C	20	D	30	A	40	D	50	A	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Leg 1 time = $60/40 = 1.5$ h; leg 2 time = $90/60 = 1.5$ h. Total 150 km in 3 h, so average = $150/3 = 50$ km/h. The plain mean of 40 and 60 gives 52.5-type values; a mis-weighted blend gives 48; 100 is just the sum of the two speeds.

2. (A) Take the path as 5 units. Time for unit 1 = $1/6$; time for the last 4 units = $4/24 = 1/6$. Total time = $1/3$ h-unit for 5 units, so average = $5/(1/3) = 15$ km/h. Mis-weighting the fifths gives 12 or 18; over-counting the fast portion gives 20.

3. (C) Multiply km/h by $5/18$: $270 \times 5/18 = 1350/18 = 75$ m/s. Multiplying by the inverted $18/5$ gives 972; dividing by 36 gives 7.5; multiplying by $18/50$ gives 97.2.

4. (B) Multiply m/s by 3.6: $35 \times 3.6 = 126$ km/h. Dividing by 3.6 gives about 9.72; halving gives 63; multiplying by 5 gives 175.

5. (D) Distance = $2 \times 5 + 1 \times 9 = 19$ km. Total time including the rest = $2 + 1 + 1 = 4$ h. Average = $19/4 = 4.75$ km/h. The plain mean of 5 and 9 gives 7; dropping the rest hour

gives $19/3 = 6.33$; 9.5 is half the distance misread as a speed.

6. (A) Period depends only on length, with T proportional to the square root of L ; mass and amplitude do not matter. So $T_P : T_Q = \sqrt{4} : \sqrt{1} = 2 : 1$. Using lengths directly gives $4 : 1$; inverting gives $1 : 2$; squaring the length ratio gives $1 : 4$.

7. (A) The climbing-and-falling needle is the speedometer (current 54 km/h); the only-increasing digits are the odometer (162 km). Time = distance/speed = $162/54 = 3$ h. Multiplying gives 108; $162/162$ gives 1; $54/162$ gives about 0.33.

8. (D) $25 \text{ m/s} \times 3.6 = 90 \text{ km/h}$, beating the train's 81 km/h. Dividing 25 by 3.6 gives about 6.9 and wrongly favours the train; the train is 81 not 90, so no tie; 25×3.6 is 90, not 108 (108 would be 30 m/s).

9. (A) Total distance = $30 + 45 = 75 \text{ cm} = 0.75 \text{ m}$. Total time including the pause = $5 + 6 + 10 = 21 \text{ s}$, so average = $0.75/21 = 0.0357 \text{ m/s}$. Dropping the pause gives $0.75/15 = 0.05$; a wrong leg total gives near 0.075; $0.75/12$ gives 0.0625.

10. (C) Slope = rise/run. First segment $36/3 = 12 \text{ m/s}$; flat segment 0; last segment $(90-36)/(12-9) = 54/3 = 18 \text{ m/s}$, the steepest. 12 is the gentler first slope; 36 is a y-value not a slope; 7.5 is the whole-trip average $90/12$.

11. (D) Equal 0.5 h intervals but growing distances 30, 45, 60 km mean speed rises, so the motion is non-uniform and speeding up. Equal intervals alone do not make it uniform when distances differ; the distances grow not shrink; straight track rules out circular.

12. (B) Up-and-down repeating motion is oscillatory and, repeating at a fixed rate, periodic. 240 strokes in 60 s means one stroke every $60/240 = 0.25 \text{ s}$. The 4 s inverts the ratio; the needle does not go in circles; a fixed 240/min rate is periodic, not non-periodic.

13. (D) Speed $P = 84/6 = 14 \text{ m/s}$; speed $Q = 48/6 = 8 \text{ m/s}$; difference = $14 - 8 = 6 \text{ m/s}$. 36 is the raw distance gap, not a speed; 2 is a wrong subtraction; 14 is P 's full speed, not the excess.

14. (C) Total time allowed = $360/90 = 4 \text{ h}$. The first leg took $270/108 = 2.5 \text{ h}$, leaving 1.5 h for the last 90 km, so required speed = $90/1.5 = 60 \text{ km/h}$. Averaging 108 and the target gives 72; 90 just repeats the trip average; using a wrong remaining time gives 120.

15. (A) Original period = $75/30 = 2.5 \text{ s}$. New period = $2.5 \times 2/5 = 1.0 \text{ s}$. In 75 s it now makes $75/1.0 = 75$ oscillations. Keeping 30 ignores the change; multiplying the period instead gives the 12-style error; using three-fifths gives 60.

16. (C) Convert to m/s: 30 m/s stays 30; $100 \text{ km/h} = 100 \times 5/18 = 27.8 \text{ m/s}$; $1620 \text{ m/min} = 1620/60 = 27 \text{ m/s}$; $0.4 \text{ km/min} = 400/60 = 6.7 \text{ m/s}$. The greatest is 30 m/s, clear of all the others.

17. (C) Since T is proportional to the square root of L , multiplying L by 9 multiplies T by $\sqrt{9} = 3$, so the period triples. Taking the factor directly gives nine times; inverting gives one-third; length does affect the period, so 'unchanged' is wrong.

18. (A) Distance from the start (a cumulative graph) rises to 50 m, stays, then the descent shows the object returning over those same 50 m. Path length = 50 m out + 50 m back = 100 m. 50 m counts only the outward trip; 0 m confuses path with displacement; 150 m wrongly adds the flat segment as travel.

19. (B) Speed = slope = $(41 - 14)/(10 - 4) = 27/6 = 4.5$ m/s. 3.5 comes from 41/10 minus a slip; 5.5 from a wrong rise; 4.1 from reading 41/10 directly instead of using both points.

20. (D) Total distance = $48 + 60 + 72 = 180$ km in 3 h, so average = 60 km/h. Because the hourly distances differ (48, 60, 72), the motion is non-uniform, not uniform. 180 is the total distance, not a speed; 55 is a miscalculated mean.

21. (C) Total distance = $48 + 48 = 96$ km; total time = $2 + 4 = 6$ h; average = $96/6 = 16$ km/h. The downstream speed is 24 and upstream 12; their plain mean 18 is the trap; 12 is only the upstream leg; 24 is only the downstream leg.

22. (B) The tip traces a circle (circular) and repeats the same path every hour (periodic). It does not swing to-and-fro, so not oscillatory; it does not move in a straight line, so not rectilinear; the motion is regular, not random.

23. (B) Since T is proportional to the square root of L , dividing T by 3 (from 6 s to 2 s) requires dividing L by 3-squared = 9. So the new length is one-ninth. One-third confuses T with L directly; one-sixth halves wrongly; three times would lengthen, not shorten.

24. (A) $75 \text{ km/h} \times 5/18 = 20.83 \text{ m/s}$, not 25, so that one is wrong. Checking the rest: $54 \times 5/18 = 15$ (correct); $90 \times 5/18 = 25$ (correct); $126 \times 5/18 = 35$ (correct). Note 90 km/h is the one that truly equals 25 m/s, so 75 km/h cannot also be 25.

25. (B) A crossing point means both lines share the same coordinates, so the objects are at the same distance from the start at that time. The slopes (speeds) generally differ, so they need not be moving at the same speed; nothing forces them to stop or reverse.

26. (B) 90 ticks in 60 s gives an interval of $60/90 = 0.67$ s between ticks. Inverting to $90/60$ gives 1.5; 90 is the tick count, not a time; 0.015 misplaces a factor of 60.

27. (C) Distance = $120 + 80 = 200$ km. Total time including the 0.5 h park = $2 + 0.5 + 1 = 3.5$ h. Average = $200/3.5 = 57.1$ km/h. Dropping the park gives $200/3 = 66.7$; the plain mean of 60 and 80 km/h leg speeds gives 60-ish; 40 is the slower leg's speed alone.

28. (D) An hourglass keeps time because sand falls through the neck at a steady rate, taking the same total time to empty. The grains do not pulse; the bulbs do not swing; it

needs no sunlight, working in the dark.

29. (B) $108 \text{ km/h} \times 5/18 = 30 \text{ m/s}$. Distance = $30 \times 75 = 2250 \text{ m}$. Using 108 m/s gives 8100; using 20 m/s gives 1500; dropping a factor of ten gives 225.

30. (A) A rising line means forward motion; an increasing steepness means the speed is growing, so the object is speeding up. Flattening (not steepening) would mean slowing down; a falling line would mean returning; a flat line would mean at rest.

31. (A) Frequency = number of oscillations / total time = $48/72 = 0.67$ per second. Inverting (time/oscillations) gives the period 1.5 s, not the frequency; 48 is the count; 72 is the time.

32. (B) Times: $40/40 = 1 \text{ h}$, $40/20 = 2 \text{ h}$, $40/60 = 0.67 \text{ h}$; total = 3.67 h for 120 km, so average = $120/3.67 = 32.7 \text{ km/h}$. The plain mean of 40, 20, 60 gives 40; a slightly different wrong mean gives 36; 30 underestimates by mis-adding the times.

33. (B) Distance = speed $\times$ time = $300000 \times 600 = 180000000 \text{ km} = 180 \text{ million km}$. 500 mistakes time for distance; 18 million drops a factor of ten; 1.8 billion adds one.

34. (C) B begins with a gentler slope (slower) but, as its upward bend steepens, its speed grows and it can overtake A. A straight line does not automatically stay ahead, nor does a curve automatically win; starting together does not keep them together once speeds differ.

35. (A) Total distance 600 m; total time = $30 + 90 = 120 \text{ s}$; average = $600/120 = 5 \text{ m/s}$. The plain mean of the leg speeds (8 m/s and 4 m/s) gives 6; 4 is only the slow leg; 8 is only the fast leg.

36. (C) Speed A = $150/10 = 15 \text{ m/s}$, speed B = $90/10 = 9 \text{ m/s}$; ratio $15 : 9 = 5 : 3$. Inverting gives $3 : 5$; $150 : 90$ does simplify (divide by 30); equal times do not make speeds equal when distances differ.

37. (B) A horizontal line means the distance stays at 60 m, so no distance is covered and the speed is zero. 60 is a y-value not a slope; 8 is the time width, not a speed; a flat line means stopped, not fast.

38. (A) Period is proportional to the square root of length, so the 16 m pendulum's period is $\sqrt{16} = 4$ times the 1 m one's. A four-times-longer period means four times fewer oscillations, so the short pendulum makes four times as many. Longer strings swing more slowly; mass is irrelevant; the factor is 4, not 16.

39. (C) The flag moves to-and-fro (oscillatory in style) but the intervals are irregular, so it is non-periodic. A clock pendulum repeats at fixed intervals, unlike this; the flag does not trace a circle; the straight pole does not make the flag's motion rectilinear.

40. (D) Distance = $150 + 90 = 240$ km. Total time including the 0.5 h idle = $1.5 + 0.5 + 1 = 3$ h. Average = $240/3 = 80$ km/h. Dropping the idle gives $240/2.5 = 96$; the plain mean of the leg speeds gives near 60-95; 120 ignores the idle and a leg.

41. (C) Out time = $4/8 = 0.5$ h; back time = $4/24 = 0.167$ h; total = 0.667 h for 8 km, so average = $8/0.667 = 12$ km/h. The plain mean of 8 and 24 gives 16; 10 and 14 come from wrong weightings of the two legs.

42. (A) 84 mm = 0.084 m; speed = $0.084/7 = 0.012$ m/s. Forgetting the mm-to-m conversion gives 12; a one-place slip gives 0.12 or 0.0012.

43. (C) Period = $50/40 = 1.25$ s. 4 minutes = 240 s, so oscillations = $240/1.25 = 192$. Using 50 s as a minute gives 160; reading 40 per minute gives 160 or 200; treating 240 s as the answer gives 240.

44. (C) First slope $40/4 = 10$ m/s; flat 0; last slope $(100-40)/(13-10) = 60/3 = 20$ m/s, the fastest. 10 is the slower first segment; 40 is a y-value not a slope; the segments are not equal.

45. (B) A water clock depends on dripping water, not the Sun, so clouds, night, and a low Sun do not stop it — those are precisely the conditions that defeat a sundial. None of the listed items disables the water clock.

46. (B) Equal distance means the quickest is fastest. Times are Q 30 min, R 36 min, P 45 min, so fastest to slowest is Q, R, P. Equal distance does not make speeds equal; the other orderings reverse or scramble the times.

47. (D) Times: $60/60 = 1$ h, $60/30 = 2$ h, $60/90 = 0.67$ h; total = 3.67 h for 180 km, so average = $180/3.67 = 49$ km/h. The plain mean of 60, 30, 90 gives 60; 55 and 45 are other mis-weighted means.

48. (A) $216 \text{ km/h} \times 5/18 = 60$ m/s. Distance = $60 \times 35 = 2100$ m. Using 216 m/s gives 7560; using 30 m/s gives 1050; dropping a factor of ten gives 210.

49. (B) A horizontal line means distance is unchanging, so the object is at rest, not moving fast — that statement is false. The other three are all true facts about distance-time graphs.

50. (A) $2.7 \text{ km} = 2700 \text{ m}$; $45 \text{ min} = 2700 \text{ s}$; speed = $2700/2700 = 1$ m/s. Multiplying by 3.6 wrongly gives 3.6; using 45 s instead of 2700 s gives 60; a factor slip gives 0.06.

51. (D) Shortening the pendulum shortens its period (T proportional to the square root of length), so it ticks more frequently and the clock gains time, running fast. A shorter pendulum swings more often, not less; period does not depend on mass; a short pendulum still swings.

52. (B) On a distance-from-start graph, a rise then a fall back to zero means the object went out to 72 m and then came back to where it began. The descent shows return, not continued speeding up, no circle, and no permanent stop.

53. (D) Speed = slope = $(52 - 16)/(8 - 2) = 36/6 = 6$ m/s. 8 comes from $16/2$; 5 from a wrong rise; 6.5 from $52/8$.

54. (D) By 11:00 the walker is $2 \text{ h} \times 5 = 10$ km ahead. The cyclist closes the 10 km gap at $15 - 5 = 10$ km/h, taking 1 h, so they meet at 12:00, by which time the cyclist has gone $15 \text{ km/h} \times 1 \text{ h} = 15$ km. The 1:00 and 11:30 options use wrong closing speeds; they do meet.

55. (D) To pass the post, the train moves its own length 120 m in 6 s, so speed = $120/6 = 20$ m/s. Multiplying gives 720; inverting parts gives 5 or 0.05.

56. (C) Both start at the origin; a steeper line means greater speed, so A pulls ahead and stays ahead at every later time. A gentler line is slower, not farther; the gap grows rather than staying fixed; since both start together, A is never behind.

57. (A) $117 \text{ km/h} \times 5/18 = 585/18 = 32.5$ m/s, not 30, so that conversion is wrong. The others check out: $36 \times 5/18 = 10$, $72 \times 5/18 = 20$, $144 \times 5/18 = 40$, all correct. (30 m/s would correspond to 108 km/h, not 117.)

58. (D) Distance = $36470 - 36200 = 270$ km. The break is part of the elapsed time, so total time = 4 h. Average = $270/4 = 67.5$ km/h. Removing the 0.5 h break gives $270/3.5 = 77.1$; 60 and 54 come from wrong distance or time figures.

59. (D) Speed = slope = $(77 - 21)/(11 - 3) = 56/8 = 7$ m/s. 7.7 comes from $77/10$; 6 and 8 from wrong rise or run values.

60. (D) From 0 to 4 s the distance rises 12 m every 2 s (6 m/s, uniform). From 4 to 6 s it rises only 6 m (3 m/s, slower). From 6 to 8 s the distance stays at 30 m, so the object is at rest. It is not uniform across all 8 s, not speeding up, and certainly not at rest the whole time since it clearly advanced first.